Teks academy

Capstone Project: Data Cleaning – Messy GDP Dataset

Dataset Provided

File: `messy_gdp_dataset.csv`

This dataset represents GDP values of different countries across years, but it has **real-world messy issues**:

- Missing values (`NaN`) in GDP.
- Inconsistent country names (extra spaces, uppercase/lowercase, abbreviations).
- GDP values stored as strings with commas (not numeric).
- Duplicate rows.
- Extra spaces and inconsistent formatting in column names.

Project Task

Your job is to **clean and prepare the dataset** so that it is ready for analysis.

Steps to Perform

1. Load and Inspect the Data

- Import the dataset using Pandas.
- Print the first 5 rows.
- Check column names, data types, and missing values.

2. Handle Missing Values

- Identify how many missing values are in the GDP column.
- Decide how to handle them (e.g., drop rows, fill with mean/median, forward fill).
- Justify your choice.

3. Fix Formatting Issues

- Clean column names (remove extra spaces, make lowercase).
- Standardize country names:
  - `" india "` → `"India"`
  - `"USA"` → `"United States"`
  - `"CHINA"` → `"China"`
  - `"germany"` → `"Germany"`
  - `"Egypt, Arab Rep."` → `"Egypt"`
  - `"Hong Kong SAR, China"` → `"Hong Kong"`

4. Convert Data Types

- Convert the GDP column from string with commas → numeric (`float`).
- Ensure `year` is of type `int`.

5. Remove Duplicates

- Find and drop duplicate rows.

6. Final Checks

- Verify there are no missing values left.
- Verify each column has correct data type:
  - `country` → string
  - `year` → integer
  - `gdp_value` → float
- Sort dataset by `country` and `year`.

7. Save the Cleaned Dataset

- Save the cleaned dataset as `cleaned_gdp_dataset.csv`.

Deliverables

1. **Python Notebook (.ipynb)** with step-by-step cleaning.
2. **Cleaned dataset (`cleaned_gdp_dataset.csv`)**.
3. **Short report (1 page)** mentioning:
   - What issues were found.
   - How missing values were handled.
   - What formatting corrections were done.
   - Why certain cleaning decisions were made.